

John Boerchers

AEROSPACE & PROPULSION · COMPUTATIONAL FLUID DYNAMICS · TURBULENT REACTING FLOWS

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Princeton, NJ

SUMMARY

Fourth-year PhD candidate at Princeton developing high-fidelity combustion models for hypersonic and supersonic propulsion. Expertise in manifold-based LES turbulence-chemistry interaction, high-speed reacting flow physics, and scientific software development (C++, Fortran, Python). Proven experience bridging fundamental research with propulsion-relevant applications, including GPU-accelerated combustion codes and collaboration with national laboratories.

TECHNICAL SKILLS

Languages: Python (proficient), Fortran (intermediate), C++ (intermediate)

HPC & Parallel: MPI / mpi4py, OpenMP, OpenACC (GPU)

Scientific Tools: NumPy, SciPy, MATLAB, Git, Bash / Linux, LaTeX

Machine Learning: PyTorch

Math Foundations: PDEs / ODEs, linear algebra, numerical methods & algorithms, optimization

EDUCATION

Princeton University

Ph.D. Candidate, Mechanical & Aerospace Engineering

2022 – May 2027 (expected)

M.A. in Mechanical & Aerospace Engineering (conferred 2024) · Graduate Certificate in Computational Science & Engineering (in progress) · Advisor: Prof. Michael E. Mueller · MAE Department Fellowship (2023)

Pennsylvania State University

B.S., Aerospace Engineering — Schreyer Honors College, Summa Cum Laude

2018 – 2022

GPA 3.99 · Honors Thesis: *A Viscous-Inviscid Channel Flow Approach for Preliminary Wind Tunnel Design* · Richard W. Leonhard Scholarship · Evan Pugh Scholar Award

RESEARCH EXPERIENCE

Computational Turbulent Reacting Flow Laboratory, Princeton University

Graduate Research Assistant — Advisor: Prof. Michael E. Mueller

2022 – Present

- Developing manifold-based combustion models that extend LES capability to high-speed reacting flows, enabling efficient high-fidelity simulations for rotating detonation engine (RDE) and scramjet propulsion research.
- Implementing and validating models within PeleC, an AMR-enabled exascale reacting flow solver, targeting rotating detonation combustor configurations.
- Collaborated with NREL to benchmark manifold models against DNS data for supersonic combustion.

Lawrence Livermore National Laboratory

Research Intern — Advisor: Boyan Lazarov

June – Aug 2022

- Built a physics-constrained neural network (PyTorch) trained on DNS data to accelerate topology optimization of porous-media flow problems — reducing design iteration cost by leveraging data-driven surrogates.

Aircraft Propulsion Research Group, Penn State

Undergraduate Researcher — Advisor: Prof. David K. Hall

May 2021 – May 2022

- Developed a viscous-inviscid coupled numerical tool for preliminary wind tunnel aerodynamic design and performance analysis.

High Pressure Combustion Laboratory, Penn State

Undergraduate Researcher — Advisor: Prof. Richard Yetter

Nov 2019 – May 2021

- Designed a CO₂ laser ignition experiment to quantify energy input and flame propagation for energetic solid propellants; contributed to published study on piezoelectric effects in nano-aluminum composites.

SELECTED PROJECTS

Princeton GPU Hackathon

June 2023

- Partnered with NVIDIA mentor to GPU-accelerate combustion chemistry kernels using OpenACC in a thermochemical solver; optimized parallel execution and GPU memory management to achieve speedup on combustion-critical code paths.

TEACHING & MENTORSHIP

Teaching Assistant, Princeton University

2024 – 2026

- MAE 426: *Rocket & Air-Breathing Propulsion Technology* – Spring 2026
- MAE 427: *Energy Conversion & the Environment* – Spring 2025 · **E-Council Teaching Award 2026**
- MAE 335: *Compressible Flows* – Fall 2024 · **Crocco Teaching Award 2025**

Undergraduate Thesis Mentor

2024 – 2025

Mentored Laura T. Thompson on RDE numerical modeling (hydrogen/methane fuel composition). Thesis earned **MAE Best Undergraduate Thesis Award** and the **SEAS Thesis Award**.

PUBLICATIONS & PRESENTATIONS (SELECTED)

A Thermodynamically Consistent Manifold Model for Premixed Deflagrations & Detonations

J.B. Boerchers, M.X. Yao, L.T. Thompson, M.E. Mueller · In Preparation, 2026

Implementation of a Thermodynamically Consistent Manifold Model for Premixed Detonations and Deflagration

J.B. Boerchers, L.T. Thompson, M.X. Yao, M.E. Mueller · APS Division of Fluid Dynamics, 2025

A Thermodynamically Consistent Manifold Model for Premixed Detonations and Deflagration

J.B. Boerchers et al. · International Conference on Numerical Combustion, 2025

Modeling Enthalpy Variations with Mixture Fraction in High-Speed Turbulent Combustion

J.B. Boerchers, M.E. Mueller · US National Combustion Meeting, 2025

Modeling and Topology Optimization of Flow Problems in Porous Media Using Neural Networks

M. Schmidt, J.B. Boerchers, B. Lazarov · ASME IMECE, 2022